

Developing Nutrition Programs for High Producing Dairy Herds

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ABSTRACT

Herd average milk production continues to increase in the US. Average milk production in Holstein herds enrolled in DHI testing programs surpassed 9000 kg in some states in 1991. Individual dairy herds have produced >14,000 kg per cow per lactation. The upper limit for milk production per cow continues to increase. A challenge exists in developing nutrition programs for these herds. The goal is to attain efficient and profitable levels of milk production while maintaining herd health and reproductive performance. Evaluation of rations currently fed to high producing herds indicate that these rations are consistent with current nutrient requirement guidelines. Many high producing herds have average DMI >4% of BW. Ration formulation principles and nutrient requirements used in development of feeding programs for high producing herds are similar to methods already in use. Optimizing DMI, optimizing rumen fermentation, and providing supplemental nutrients are key factors in meeting tissue nutrient demands in this formulation process.

(Key words: nutrition programs, milk production, dry matter intake, nutrient requirements)

INTRODUCTION

Average milk produced per cow per lactation continues to increase in the US. Milk production for DHI herds increased from 4160 to 8178 kg/yr per cow between 1950 and 1990. This continued improvement in productivity is a combination of genetic and environmental

factors. The average milk production for cows on DHI has surpassed 9000 kg in some states. Individual dairy herds are producing >14,000 kg of milk/yr per cow.

Development of feeding programs for these high producing herds can be a challenge. Dairy herd managers expect and demand nutrition programs that support high milk production while they still control feed cost per unit of milk produced. At the same time, cow health and reproductive performance parameters must be considered in the formulation process.

Traditionally, nutrition programming has been based on nutrient requirements derived from research trials. This system has worked well and continues to evolve. The majority of this research database is from cows producing <9000 kg/yr. Can this requirement database be used to formulate rations for herds producing ≥13,000 kg/yr?

HIGH PRODUCING COWS

What is the definition of high production? No set criteria can provide an answer. Individual dairy herd managers have their own definitions. This symposium has been targeted at herds producing >13,600 kg/yr. McCullough (32) has defined a super cow as producing ≥20 kg of milk/kg of BW. A summary of information relative to individual Holstein cows producing 18,144 kg has been presented (1). These cows produced between 21 and 32 kg of milk/kg of BW. Review papers pertaining to high producing cows have been presented (5, 12). The nutrition programs for high producing herds in Wisconsin have also been summarized (W. T. Howard and R. D. Shaver, 1991, personal communication). The information from these herds and cows can provide a basis for examination of the approaches to developing nutrition programs.

If a dairy herd is to attain a 13,600-kg average, average daily milk production needs to be 37.3 kg per cow, assuming that the herd

Received August 1, 1992.

Accepted January 6, 1993.

has 100% DIM. Daily milk production per milking cow needs to be 43.9 or 41.4 kg for herds with 85 or 90% DIM. An additional parameter to evaluate is peak milk production needed to attain a specified herd average. Table 1 shows peak production and persistency for dairy cows for four ranges of herd averages. Peak milk production is greater in higher producing herds. The decline in persistency is greater in the higher producing herds.

DIM

The most important variable influencing cow productivity is DIM (33, 40). A survey of high producing dairy cows in England and Wales concluded that high DIM was a key management attribute in high producing cows (35, 46). Several equations have been developed to predict DIM of lactating dairy cows (27, 34, 42), but these equations have not been validated with high producing dairy cows. The DIM attained by the cow Ellen varied from 4.4 to 6.7% of BW over the lactation (1). During her highest lactation, she produced 25,248 kg of milk. Limited quantitative measurements of actual DIM exist for cows producing >12,000 kg of milk per lactation.

The rate of increase in DIM during early lactation is a primary factor that determines energy intake and energy balance. The 1989 NRC (34) suggests that DIM may be up to 18% less in early lactation than would be predicted at maximum intake. Weekly adjustment factors for DIM have been reported (27, 42). The DIM during wk 1 of lactation was 67% of the maximum attained at 8 wk post-calving (42). If the rate of concentrate feeding is increased too rapidly postcalving, the potential exists to depress forage intake in the early lactation cow.

Does milk production drive DIM or does DIM drive milk production? This question is frequently discussed and debated. However, information from bST trials provides at least a partial answer. Bauman et al. (3) reported that milk production responses occurred within days of initiation of bST treatment. The DIM did not respond until 4 to 6 wk after initiation of bST treatment, providing some evidence that milk production drives DIM. The slower increase in DIM than milk production post-calving may also provide a partial answer to this question.

A critical interrelationship exists between energy demand, energy intake, and energy balance in the high producing dairy cow (2). This interrelationship is most critical during early lactation when reproduction patterns are established. Change in BW has been often used as an index of energy balance or status. The interpretation of BW change in early lactation cows is confounded with increases in rumen fill that occur as intake increases (15, 27, 38, 45). The use of a body condition scoring system may be an alternative method to evaluate energy balance. A 1-unit change in body condition score in Holstein cows represented a change of 56 kg of BW (38). This relationship should assist in refining energy balance calculations in early lactation dairy cows. Net energy deficits were >20 Mcal/d in high producing, early lactation cows (2).

The energy value of feeds is also altered by DIM. A positive relationship exists between DIM and rate of passage of feed through the cow (13, 14, 33, 39, 40). The depression in digestibility that occurs as intake increases is not uniform for all feed fractions, making uniform adjustment of energy values for intake level difficult. Computer modeling approaches

TABLE 1. Peak milk production and persistency postpeak at four herd averages.¹

Herd average (kg per cow)	Lactation 1		Lactation 2		Lactation ≥3	
	Milk	Persistency	Milk	Persistency	Milk	Persistency
	(kg)	(kg/mo)	(kg)	(kg/mo)	(kg)	(kg/mo)
7250 to 7711	25.4	-.9	32.2	-1.8	34.8	-2.1
9072 to 9525	30.2	-.94	38.8	-2.1	42.0	-2.4
10,886 to 11,340	35.9	-1.08	46.1	-2.55	49.2	-2.8
12,700 to 13,154	43.7	-1.51	55.4	-3.25	58.7	-3.5

¹L. R. Jones, 1992, personal communication.

may be needed to make this adjustment more systematically and dynamically (16, 33). Nutrient digestibility may also be higher in cows fed mixed rations than when the same components are fed separately (21, 24), suggesting that feeding frequency, meal size, and pattern of intake can influence digestibility of feeds.

FORAGE QUALITY

Forages are the foundation of dairy ration formulation. The cell-wall components of forage are related to potential forage intake (26, 40). The rate of rumen degradation is also variable within and between forage species. Kawas et al. (26) reported that, during wk 10 to 26 of lactation, dietary fiber should be <20% ADF or 29% NDF. These percentages were needed by cows producing >35 kg/d of 4% FCM to maintain positive body tissue balance. Several studies (25, 26, 29, 30, 39) using alfalfa forages have examined the concepts of forage quality, grain supplementation, and milk production. Rate of fiber digestion was faster in early vegetative hay than in late bud or full bloom hay (29). Milk production did not differ when rations were formulated with similar diet fiber contents using three qualities of alfalfa (25) or when midlactation cows were fed rations with high quality orchardgrass or high quality alfalfa (44). A forage source that is high in intake potential and digestibility is still the key to development of rations to support high milk production.

NUTRIENT UTILIZATION

The nutrients consumed by the dairy cow are utilized to meet metabolic demands for maintenance, milk production, growth, and reproduction. The mechanism by which nutrient intake is partitioned to meet these various demands is not fully understood (2). In addition to nutrient demands to support these physiological functions, metabolite fluxes of lipid, protein, carbohydrates, and minerals must be considered (2, 4, 47).

As milk production and DMI increase, a larger proportion of the total nutrient intake is available to support milk production. Figure 1 shows the proportion of the total energy and protein intake that is available for milk production in dairy cows at various milk productions.

For dairy cows producing 60 kg/d of milk, approximately 91% of the CP intake is available to support milk production. The corresponding figure for net energy intake is 80%.

An additional consideration is the quantity of each of the various metabolites that must be available for the high producing dairy cow (2, 4, 6, 28, 31, 47). A dairy cow producing 89 kg/d of milk has been estimated to require >7 kg/d of glucose (47). The quantity of glucose that can be provided by absorption from the digestive tract has been indicated to be ≤ 1 kg/d (2), suggesting that a dramatic increase in gluconeogenesis in early lactation is necessary for high milk production. Precursors needed to support this increased gluconeogenesis include propionate, lactate, amino acids, and glycerol. Similar metabolic adjustments in amino acid, lipid, and mineral metabolism are also required in the early lactation cow.

RUMEN FERMENTATION

Nutrient utilization and microbial protein synthesis in the rumen are important for the design of efficient, profitable rations. Review articles (8, 10, 36, 37) have examined carbohydrate and protein considerations in optimizing rumen fermentation. One limitation for fiber utilization in the rumen appears to be rumen pH (17, 22). Decreasing rumen pH below 6.0 to 6.2 appears to decrease the rate and the extent of fiber utilization. Feeding strategies

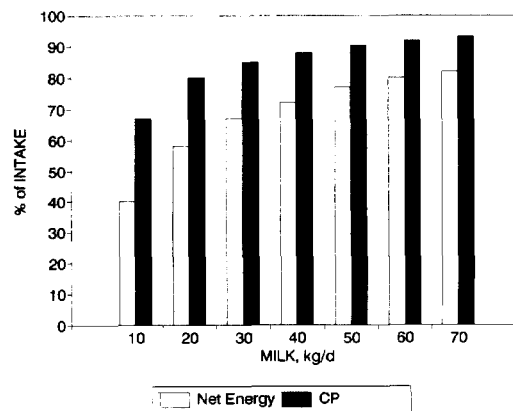


Figure 1. Percentage of total nutrient intake utilized for milk production.

need to be designed to minimize fluctuations in rumen pH.

The potential benefit of providing protein and carbohydrate sources with similar rumen degradation rates has been examined (7, 18, 19, 23, 37, 43). Weiss et al. (43) reported that dairy cows fed corn grain rather than barley had a higher milk fat test and produced more 4% FCM. Herrera-Saldana et al. (18) found that microbial N synthesis was higher in barley-based rations than with milo. This finding is at least partially related to a greater rate and extent of carbohydrate in the rumen from barley.

The estimation of rumen degradation rates of carbohydrates and proteins is essential to develop a database for ration formulation. The 1989 NRC (34) publication has a limited listing of the rumen undegradability of protein in feedstuffs. Many of these values represent only a limited number of observations. No currently accepted, standardized method exists to determine the undegradability of feeds. However, these values should not be assumed to be constant. The assumption should be that degradation rates change with intake level, rumen environment, and liquid and particulate passage rate. This issue has been examined, and some preliminary values are available that reflect passage rates (9, 20, 36). Design of post-ruminal supplementation programs to optimize performance will continue to be difficult until a better understanding and accounting of the rumen contribution to tissue demands can be attained. The amino acid profile of the undegraded protein fraction reaching the small intestine can be altered by manipulation of ration ingredients (8, 11, 41). The challenge is to know when this approach is both necessary and economical. Presently, computer models provide one approach (16, 33). The variation in results in the literature when amino acids are added to the ration is at least partially due to the inability to predict rumen microbial amino acid production.

RATION FORMULATION

How can the concepts mentioned herein be used to provide guidelines to develop rations for high producing herds? The same approaches, thoughts, and concepts commonly used in ration formulation should apply. Dairy

cows treated with bST altered nutrient partitioning in favor of milk production (3). However, the maintenance requirement and dietary digestibility were not changed. The nutrient requirements for the higher milk production can be determined using the NRC (34) publication.

A second method is to evaluate feeding programs of high producing dairy herds and to compare them with NRC (34) guidelines. Although limited, field data has value when used in the proper context. One data set, currently available from University of Wisconsin (W. T. Howard and R. D. Shaver, 1991, personal communication), summarizes the feeding programs of seven Holstein herds averaging 12,602 kg per cow in 1990. The average ration nutrient density for NE_L in these herds was 1.74 Mcal/kg. Similar densities for CP, ADF, and NDF were 19.3, 19.2, and 28.2%. The rumen-undegradable protein, expressed as a percentage of CP, was 36%. These nutrient densities are similar to that in the NRC (34) nutrient requirement publication.

The following procedures and guidelines are presented as a basis for development of rations for high producing herds. 1) The goal of feeding management is a continuous supply of fresh, palatable, and high quality feed. The feeding system used is not as important as the number of meals that the cow is enticed to consume. A wide variety of feeding systems are used in high producing herds. 2) The key factor that determines success or failure of a feeding system is DMI. If peak milk production ≥ 55 kg/d is to be attained, DMI must be ≥ 27 kg/d for the cow to be in energy balance, which is 4.25% of BW for a 635-kg cow. Individual cows likely will approach DMI of 5% of BW. The average DMI in the Wisconsin herds was 25 kg, which represents the average for the herd as a whole, not only for cows at peak production. The DMI of some individual cows in these herds can be expected to be >30 kg/d. Figures 2 and 3 show the milk production that can be supported with varying nutrient densities of NE_L and CP at different amounts of DMI. 3) The quantities, balance, and types of protein and carbohydrate sources need to be considered for optimization of rumen fermentation. An imbalance of protein and carbohydrate sources can depress microbial protein synthesis (18, 23, 37). The

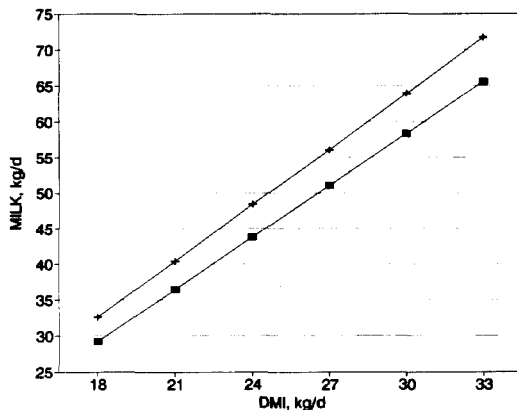


Figure 2. Projected milk production at varying DMI with two ration energy concentrations: NE_L at 1.67 Mcal/kg (■) and 1.8 Mcal/kg (+).

NRC 1989 (34) protein system estimates the quantities of rumen-degraded and rumen-undegraded protein required. The soluble protein content should also be evaluated and maintained at about 50% of the rumen-degraded protein. Soluble protein content must be addressed when the NRC (34) protein model is used. One portion of this model estimates microbial protein synthesis based on NE_L or TDN content of the feeds. If total ration fat exceeds 3.5% of ration DM, the NE_L or TDN value needs to be adjusted before microbial

protein synthesis is estimated; otherwise, the quantity of rumen-undegraded protein needed in the ration will be underestimated. Microbial protein synthesis can also be increased by controlling the ratio between rumen-degraded protein and nonstructural carbohydrates (23). The current NRC guidelines (34) for energy, fiber, and protein appear to be sufficient for use in ration development for high producing herds. The energy, fiber, and protein components of the rations fed to the Wisconsin herds and other high producing herds conform quite well to NRC (34) guidelines. 4) Additional post-ruminal supplementation of protein and energy is needed to meet tissue demands for milk synthesis in high producing cows. This supplementation can be achieved in rations only by prior determination of tissue needs and subtraction of rumen contribution. Computer models represent a logical approach (16). Current computer models can be used effectively to define research opportunities. 5) Feed additives, including added fat, may be justified in specific situations. The key is to define a rationale and justification for incorporation of additives into the ration. Daily feed cost can be increased significantly when additives are used. The number and type of feed additives used in high producing herds vary considerably. Attaining high DMI and optimizing rumen fermentation may decrease the need to incorporate some feed additives in rations. An individual evaluation is needed for each farm or group of cows. Selective use of feed additives may be justified in high producing herds.

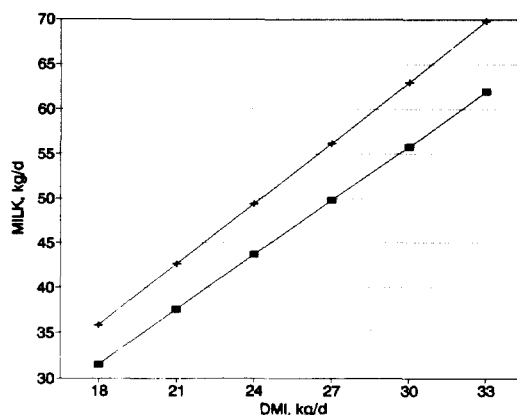


Figure 3. Projected milk production at varying DMI with two ration protein concentrations: 17% CP (■) and 19% CP (+).

CONCLUSIONS

The principles and procedures used to develop feeding programs for high producing herds exist within the framework of the current NRC (34) requirements. Evaluations of feeding programs currently utilized by high producing herds indicate that energy, fiber, protein, and mineral components of the rations are similar to those in current requirement guidelines. The primary difference appears to be higher DMI. Development of feeding programs for high producing herds should be approached in steps. The first step is to develop a feeding management system to attain high DMI, which requires the availability of high quality, rapidly digestible forages. The second step is to bal-

ance the carbohydrate and protein fractions in the ration to optimize rumen fermentation and microbial protein synthesis. The final step is to provide supplemental nutrients to the intestine to ensure that tissue nutrient needs are met. The end result of this approach will be efficient and economical rations that support high milk production.

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